



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
SEMESTER 1 2024/2025

SECP2523 Database

Section 01

Assignment 1

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Question 1

a.

medID, patientID, date → refillAllowed, PatientName, dosage, medName, shelfLife,
doctorID, doctorName
medID → medName, shelfLife
patientID → PatientName
doctorID → doctorName

b.

Patient (medID, patientID, date, refillAllowed, dosage, doctorID, doctorName)
PK: {medID, patientID, date}
FK: medID REFERENCES Medicine (medID)
FK: patientID REFERENCES PatientInfo (patientID)

Medicine (medID, medName, shelfLife)
PK: medID

PatientInfo (patientID, PatientName)
PK: patientID

c.

Patient (medID, patientID, date, refillAllowed, dosage, doctorID)
PK: {medID, patientID, date}
FK: medID REFERENCES Medicine (medID)
FK: patientID REFERENCES PatientInfo (patientID)
FK: doctorID REFERENCES Doctor (doctorID)

Medicine (medID, medName, shelfLife)
PK: medID

PatientInfo (patientID, PatientName)
PK: patientID

Doctor (doctorID, doctorName)
PK: doctorID

Question 2

a.

FD1: $A, B, C \rightarrow D, E, F, G, H, I$

FD2: $C \rightarrow D$

FD3: $A \rightarrow E, F, G$

b.

ABC (A, B, C, H, I)

PK: {A, B, C}

FK: A REFERENCES A (A)

FK: C REFERENCES C (C)

A (A, E, F, G)

PK: A

C (C, D)

PK: C

c.

ABC (A, B, C, H, I)

PK: {A, B, C}

FK: A REFERENCES A (A)

FK: C REFERENCES C (C)

A (A, E, F, G)

PK: A

C (C, D)

PK: C

Question 3

a.

$M, N \rightarrow O, P, Q, R, S, T, U$

$N \rightarrow Q$

$O \rightarrow P$

$U \rightarrow T$

b.

$MNQ (\underline{M}, \underline{N}, O, P, R, S, T, U)$

PK: $\{M, N\}$

FK: N REFERENCES N(N)

$N (\underline{N}, Q)$

PK: N

c.

$MNQ (\underline{M}, \underline{N}, \underline{O}, R, S, \underline{U})$

PK: $\{M, N\}$

FK: N REFERENCES N(N)

FK: O REFERENCES O(O)

FK: U REFERENCES U(U)

$N (\underline{N}, Q)$

PK: N

$O (\underline{O}, P)$

PK: O

$U (\underline{U}, T)$

PK: O

Question 4

a.

PropertyID $\rightarrow$ County, LotNo, Area, Price, TaxRate

County, LotNo $\rightarrow$ Area, Price, TaxRate

County, Area $\rightarrow$ Price

County $\rightarrow$ TaxRate

b.

Second Normal Form. Because every non primary-key attribute is fully functionally dependent on the primary key (PropertyID).

c.

PropertyID (PropertyID, County, LotNo)

PK: PropertyID

FK: {County, LotNo} REFERENCE Lot ({County, LotNo})

FK: County REFERENCES CountyTaxRate (County)

Lot (County, LotNo, Area, Price)

PK: {County, LotNo}

CountyTaxRate (County, TaxRate)

PK: County